

Exam Date & Time: 04-Dec-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov / Dec -2024

Reinforcement Learning [CSE-4028]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Compare and Contrast Reinforcement Learning and Supervised Learning. (5)
 - A)
 - B) Classify the types of Reinforcement learning and explain the elements involved in Reinforcement learning. (5)
- 2) Suppose that we have the following information:
There is 60% of chance that will rain today, 50% chance of that will rain tomorrow and 30% of chance that does not rain either day.
 - A) Assuming above data find the following probabilities (5)
 - a) The probability that it will rain today or tomorrow.
 - b) The probability that it will rain today and tomorrow.
 - c) The probability that it will rain today but not tomorrow.
 - d) The probability that it either will rain today or tomorrow but not both.
 - B) In a presidential election, there are four candidates. Call them A, B, C, and D. Based on our polling analysis, we estimate that A has a 20 percent chance of winning the election, while B has a 40 percent chance of winning. What is the probability that A or B win the election? (3)
 - C) **Prove $P(A^c) = 1 - P(A)$** (2)
- 3) You are a marketing analyst for a mobile app that offers various features to enhance user experience. The app currently provides different promotional incentives to encourage users to engage more with specific features. The promotions include Incentive A (free premium feature for one month), Incentive B (bonus points for in-app purchases), Incentive C (a one-time discount on subscription), and Incentive D (access to exclusive content). Each promotional incentive has an unknown effectiveness in driving user engagement, and your goal is to determine which incentive results in the highest user interaction. To achieve this, you decide to apply the Upper Confidence Bound (UCB) algorithm to optimize which promotion to show to users. Discuss the steps involved in calculating the upper confidence bounds for each incentive and the criteria for selecting which incentive to show. (5)
 - A)

- B) Use the above scenario in question 3(a), Discuss how the UCB algorithm helps to balance exploring different promotional incentives (to learn about their effectiveness) and exploiting the best-performing incentive (to maximize user engagement). (5)
- 4) Explain the concept of the agent-environment interface in AI and discuss how it enables an AI system to perceive, interact and adapt to its surroundings. (5)
- A)
- B) Consider a robot in a maze-navigation task. The agent robot aims to find the maze's exit to earn a positive reward. Explain the agent environment perception, action, and feedback for the mentioned scenario in the given question. (5)
- 5) Explain the Monte Carlo method for policy evaluation in Reinforcement Learning. Describe its methodology in detail. (5)
- A)
- B) Discuss the role of Actor critic methods in reinforcement learning. (5)

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